

Venture Capital Syndication: Synthesis and future directions

Mikko Jääskeläinen

Aalto University, PO Box 15500, FI-00076 Aalto, Finland

Corresponding author email: mikko.jaaskelainen@aalto.fi

This paper reviews and synthesizes the extant literature on venture capital syndication. By considering the questions of how, why and when syndication affects the performance of venture capitalist (VC) firms and their portfolio ventures, a schematic structure of the syndication literature is formed and areas for further research are identified. The results of the review show that, while the venture-level aspects are relatively well understood, the current literature lacks an understanding of how and why syndication affects the performance of VC firms. This suggests that more attention should be directed towards syndication as a component of the overall strategy of VC firms.

Introduction

The frequent and continued co-operation of venture capitalists (VCs) through the syndication of investments is one of the defining features of the VC industry. Instead of investing in new ventures alone, VC firms form syndicates in which multiple investors provide financing for a venture.¹ Syndicates are a form of interorganizational co-operation which serve the purposes of financial intermediation, as well as the goals of individual VC firms. On the one hand, syndicates have formal contractual structures that define the terms of investment between participating VC firms and entrepreneurs and, on the other hand, the decision to form and participate in a syndicate is driven by the needs and opportunities of both the ventures being financed and the VC firms

themselves. Accordingly, the syndication of investments is an inherently multilevel phenomenon combining aspects of contracting, venture development, VC firm strategies, partnership formation and inter-organizational networks.

Consequently, in addition to attracting research interest in its own right, syndication has provided a rich empirical setting for an increasing number of studies examining the dynamics of interorganizational relationships and their effects (e.g. Echols and Tsai 2005; Hochberg *et al.* 2007; Kogut *et al.* 2007; Podolny 2001; Sorenson and Stuart 2008, 2001). For this review, we identified over 80 syndication-related articles published in finance, economics, sociology, entrepreneurship, strategy and management-related journals. However, this cumulative interest in syndication has resulted in a fairly fragmented view. While reviews exist of specific aspects of syndication, such as contracting (Tykvova 2007), motivations (e.g. Lockett and Wright 2001; Manigart *et al.* 2006) and the strategic approaches of VC firms (De Clercq and Dimov 2010), these contributions have focused on their specific areas, and the understanding of aspects covered by the research as a whole is dispersed among individual contributions. Consequently, the research lacks an integrated view on syndication that would allow us to assess what is known and where further contributions are needed.

¹Approximately 40–80% (Jääskeläinen *et al.* 2006; Manigart *et al.* 2006; Wright and Lockett 2003) of all investments made by venture capitalists are syndicated. The share of syndicated investments in all investments varies with respect to markets and years. While in the UK in 2001, only 13.6% of investments were syndicated (Manigart *et al.* 2006; Wright and Lockett 2003), the typical share for European markets is between 40% and 50% (Wright and Lockett 2003), and the corresponding figure for US markets is closer to 80% (Jääskeläinen *et al.* 2006).

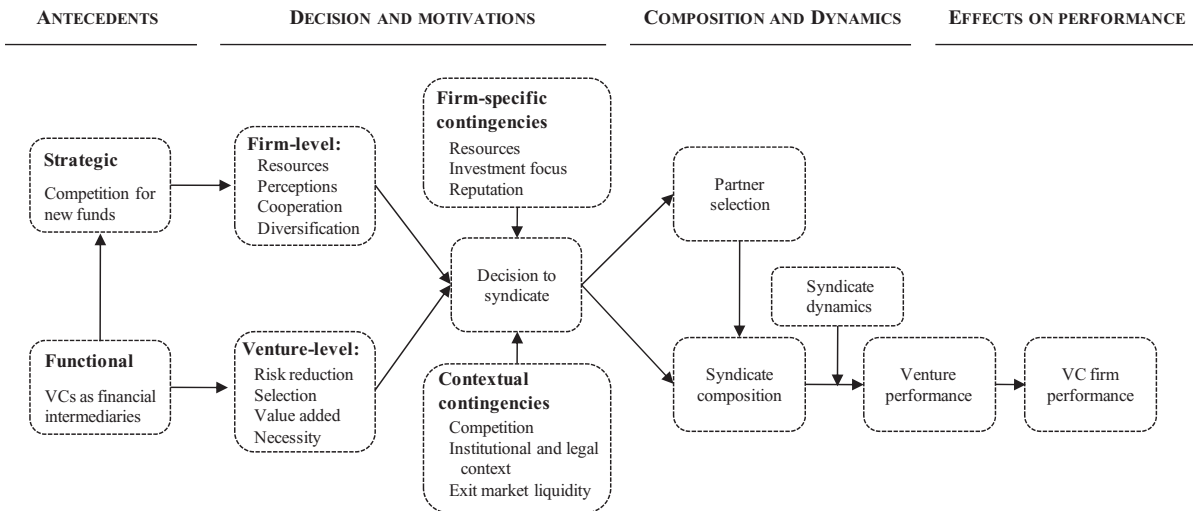


Figure 1. Schematic structure of syndication process based on extant literature

Accordingly, we review the existing literature both to identify areas for further contributions and to facilitate the further use of syndication as an empirical context by providing an accessible description of the motives, contingencies and outcomes of syndication. To structure the review and to illustrate the aspects of syndication, we present the following questions to the literature: How, why and under what circumstances does syndication influence the performance of ventures and VC firms? We review the published literature and selected contemporary working papers on venture capital syndication² for the purpose of providing the answers to these questions when they can be found in existing research and identifying the gaps when we find the research lacking.

The key elements of the review are presented in Figure 1, which provides a schematic structure of syndication – from its antecedents to performance effects – as presented in the extant literature. The review follows the elements of Figure 1 in five steps.

²It should be noted that we specifically focus on literature on venture capital and private equity syndication, thus removing from our scope the research addressing syndication in other financial settings, such as investment banking and loan markets. While sharing the aspects of information production, joint decision-making and risk-sharing, the syndicates formed around securities offerings are significantly different from venture capital syndication. Most importantly, syndicates formed to facilitate issues or loans are essentially focused on short-term information and liquidity production (e.g. Pichler and Wilhelm 2001) and thus lack one of the defining issues of venture capital syndication, that is, long-term value creation. For a concise review of investment banking syndicates, see, e.g. Song (2004).

First, we trace the antecedents of syndication to the role of the VC as a financial intermediary, identifying both a functional and strategic antecedent. Second, we review the literature focusing on the decision to syndicate, categorizing the suggested drivers into firm- and venture-level motivations. Third, we address the syndicates in terms of their structure, composition and dynamics. Fourth, we review the literature addressing the performance effects, and finally we conclude with suggestions for further research.

Functional and strategic antecedents of syndication

VCs as financial intermediaries and strategic actors

The reasons why VCs syndicate their investments can be addressed on two levels. The existing research on syndication motivation has focused on identifying the direct effects that VCs seek from syndication. These effects have been found to fall into categories such as risk reduction, deal flow generation, deal selection and value-adding (De Clercq and Dimov 2010; Lockett and Wright 2001; Manigart *et al.* 2006). However, we can step back and ask why VCs seek these effects, i.e. what the causes and antecedents of syndication are. This shifts our focus to the drivers of the syndication, which are reflected not only in the decision to syndicate, but also in the structure, composition and effects of syndicates. We suggest that the aspects of syndication can be fruitfully addressed from two complementary

antecedents: (1) the VC firm's role as a financial intermediary, here referred to as the 'functional antecedent'; and (2) its strategic needs and opportunities to survive and perform within the frames of this role, referred to as the 'strategic antecedent'.

The functional antecedent stems from the role of VCs as financial intermediaries. The economic reason for the existence of the venture capital industry and the value of VCs as financial intermediaries is based on the contractual structure of the industry (Black and Gilson 1998; Sahlman 1990), which solves the otherwise prohibitive information problems stemming from moral hazards and asymmetric information related to the financing of new ventures (Amit *et al.* 1998; Chan 1983; Diamond 1984; Gompers 1995; Kaplan and Stromberg 2003; Leland and Pyle 1977; Ueda 2004). To solve the agency and information concerns between ventures and VC firms, VCs use contracts that, on the one hand, screen ventures *ex ante*, tying the entrepreneur's effort to the allocation of control and cash flow rights (Dessi 2005; Kaplan and Stromberg 2003), and, on the other hand, provide VC firms with effective means for *ex post* monitoring and involvement through board memberships and the staging of the investments (Gompers 1995; Sahlman 1990). In addition to using effective contracting, VCs specialize in and focus their operations on specific industries (Gupta and Sapienza 1992), financing stages (Norton and Tenenbaum 1993) and geographical regions (Lerner 1995) to enhance the screening and monitoring of ventures.

This role of VCs as informed intermediaries is also the basis for the syndication of investments. Joint effort in the selection of investments leads to enhanced screening (Brander *et al.* 2002; Casamatta and Haritchabalet 2007; Cestone *et al.* 2007; Cumming 2006a; Cumming *et al.* 2010; Dimov and Milanov 2010; Lerner 1994) and facilitates the monitoring of the ventures (Fritsch and Schilder 2008; Lerner 1995; Meuleman *et al.* 2009; Sorenson and Stuart 2001). In other words, from the perspective of financial intermediation, VCs syndicate and structure their syndicates in order to increase the amount of information, skills and resources available for the decision-making, monitoring and development of individual ventures. This enhances the value of the investment by reducing the costs of asymmetric information and agency and increasing the size and probability of positive outcomes.

The strategic antecedent is based on the strategic behaviour of VCs in their role as financial intermediaries. While the contractual structure of venture

capital mitigates the inherent uncertainty, asymmetric information and agency costs of investing in new ventures, the same conflicting interests exist between VC firms and their investors. To create credible incentives for VC firms to screen, manage and monitor ventures to maximize the returns on capital invested by institutional investors, the lifetime of funds is limited and the compensation of VC firms is based on their performance (Gompers and Lerner 1999; Sahlman 1990). For VC firms, this creates a situation not unlike the staging of investments, which incentivizes ventures to perform in order to secure further financing. The success of VCs in the process of raising further funds to continue their operations is dependent on their ability to generate a realized return which compares favourably with that of their competitors. Accordingly, from the perspective of the strategic antecedent, VCs syndicate and structure their syndicates to maximize both their absolute and relative performance, as implied by the incentive structures and competition for further funds, respectively.

Applicability of the two antecedents

In broad terms, the two antecedents appear to imply largely identical consequences: VCs syndicate in order to increase their performance by acquiring access to information, skills and resources in order to screen and develop ventures. However, the goals are not perfectly aligned. To illustrate the differences between the two antecedents and to provide an example of how they offer an effective approach to analysing the motives, consider the oft-cited motive of the diversification of a portfolio (e.g. Bygrave 1987; Checkley 2009; Kaiser and Lauterbach 2007; Lerner 1994; Lockett and Wright 2001). One view in the extant research has addressed the diversification of the portfolio as a financial motivation, considering it as a means to reduce the variability of portfolio returns and thus emphasizing diversification as a goal in itself (Kaiser and Lauterbach 2007; Lerner 1994; Lockett and Wright 1999, 2001; Manigart *et al.* 2006). However, as noted by Checkley (2009), diversification on the fund level may even increase the risks of limited partners (LPs) on the aggregate level. Therefore, another view on diversification has suggested that diversification reduces the risks of producing low results that might compromise the VC firm's abilities to attract further funds (Lerner 1994; Lockett and Wright 2001). That is, when considered from the perspectives of functional and strategic antecedents, the motive for syndication with the

purpose of diversification seems to stem more from strategic motives than functional ones.

Thus, the two antecedents both allow us to structure the research on the motivations and forms of syndication and lead us to ask whose performance and what type of performance is relevant when considering syndication. While the role of VCs as financial intermediaries provides the general framework in which the firms operate and thus that perspective provides partial explanations for syndication, the strategic interests of VCs generate the second set of explanations that are able to account for phenomena such as diversification, window-dressing (Lerner 1994) and the use of syndication as an entry barrier (Hochberg *et al.* 2010). In essence, the two antecedents help us to untangle the multiple levels of analysis and interests, and detail the drivers that enable, force and motivate VCs to syndicate.

Syndication: decision and motivations

Motives and contingencies identified by extant research

What, then, are the motives presented by the existing literature and how does the adoption of the functional and strategic antecedents help us to structure this

literature? As the extant literature has focused on the effects rather than the causes of syndication, the functional and strategic antecedents are at least partly inseparable within the motives presented by the extant research. Therefore, rather than forcing the extant research into categories based on the two antecedents, we classify the motives on the basis of whether they are motivated by the performance and opportunities of the venture or of the VC firm (presented in Table 1). Nevertheless, we posit that the functional and strategic antecedents are both prominent when discussing the motives on the level of ventures and VC firms and, accordingly, we highlight the roles of the two antecedents within these categories. While it is possible to present such a preliminary analysis, there is need for further research in order to disentangle the consequences of antecedents and their relative importance in each case.

Venture-related motivations

The extant literature presents two main categories for syndication motivations related to ventures: (1) the necessity dictated by the information and agency concerns and (2) needs related to the characteristics of the venture. First, regarding necessity, studies on contracting have suggested that syndication may solve concerns about asymmetric information and

Table 1. Motivations to syndicate on VC firm, portfolio and deal levels

Firm-level motivations	
Leveraging existing/compensating for lacking resources	Deal flow: Bygrave (1987); Manigart <i>et al.</i> (2006); Fritsch and Schilder (2008) Selection expertise and capabilities: Casamatta and Haritchabalet (2007); Dimov and Milanov (2010) Value-adding capabilities: Jääskeläinen <i>et al.</i> (2006); Manigart <i>et al.</i> (2006); Dimov <i>et al.</i> (2007); De Clercq <i>et al.</i> (2008); Dimov and Milanov (2010); Deli and Santhanakrishnan (2010), Verwaal <i>et al.</i> (2010), Dal-Pont Legrand and Pomet (2010); Hopp (2010a, 2010b; Hopp and Rieder, 2011) Market-specific knowledge: Mäkelä and Maula (2008); Meuleman <i>et al.</i> (2009) Financial resources. Ferrary (2010); Gottschalg and Gerasymenko (2008)*
Managing perceptions of the VC firm	Reputation-building: Lerner (1994) Structural positioning: Milanov and Shepherd (2008)*
Managing interorganizational relationships	Entry deterrence: Hochberg <i>et al.</i> (2010) Networks: Castilla (2003); Fund <i>et al.</i> (2008); Keil <i>et al.</i> (2010),
Managing portfolio	Reducing risk of underperforming peers: Lerner (1994); Lockett and Wright (2001) Diversification: Lerner (1994); Lockett and Wright (1999) Lockett and Wright (2001); Manigart <i>et al.</i> (2006); Kaiser and Lauterbach (2007)
Deal-level motivations	
Venture-related factors	Selection: Brander <i>et al.</i> (2002); Cumming (2006a); Dimov and Milanov (2010); Cestone <i>et al.</i> (2007)* Value added: Brander <i>et al.</i> (2002); Manigart <i>et al.</i> (2006); Dimov and Milanov (2010) Risk reduction: Manigart <i>et al.</i> (2006)
Necessity	Asymmetric information between VCs: Admati and Pfleiderer (1994); Lerner (1994); Fluck <i>et al.</i> (2009)* Asymmetric information between VCs and ventures: Hellmann (2002); Huang and Xu (2003); Schmidt (2003); Bachmann <i>et al.</i> (2006)*; Fluck <i>et al.</i> (2009)*

*Marks the contributions of emerging literature.

moral hazards between VCs and ventures. Syndication, or a commitment to syndication, is seen as a necessary mechanism for ensuring a credible commitment to both the continuation and abandonment of ventures (Admati and Pfleiderer 1994; Fluck *et al.* 2009; Huang and Xu 2003), as well as a required vehicle for protecting intellectual property (Bachmann and Schindele 2006) and aligning the interests of financial and strategic investors (Hellmann 2002). The second venture-related set of motivations stems from the characteristics of the venture in question. Depending on the resources of the VC firm, it may need to: (1) resort to evaluations of other VCs to ensure a robust selection; (2) access the expertise and contacts of other VCs in order to augment its own resources to ensure sufficient contributions to the development of the venture; and (3) limit its exposure to the venture-specific financial risk by reducing its share of the required investments.

In terms of the functional and strategic antecedents, the two sets of venture-level motives appear more prominently functional than strategic ones. Syndication as a necessity relates directly to the functional antecedent, as from this perspective the syndication facilitates the mitigation of the inherent agency and information concerns related to financial intermediation. In addition, in terms of enhancing the selection, value added and risk reduction in individual investments, the motives can be interpreted to reflect functional antecedents as they serve to enhance the accuracy and the outcomes of individual investments and thus the functioning of the intermediation. However, it should be noted that the venture-specific motives can alternatively be seen as reflecting the strategic antecedents if syndication is used continuously to compensate for a lack of selection and value-adding capabilities. In this case, the motives are perhaps more accurately described as stemming from strategic concerns on the VC firm level than on the venture level.

VC firm-related motivations

On the VC firm level, motivations stem from needs and opportunities that are not directly related to the specific characteristics of the venture in question. That is, VCs use syndication as part of their overall investment strategy, and it serves as a means to enhance the performance and the chances of survival of the firm. The motivations addressed by the literature can be roughly classified into three broad categories, i.e. motivations related to: (1) resource

leverage and acquisition; (2) the management of perceptions and interorganizational relationships; and (3) fund-level risk management. In the following paragraphs, attention is concentrated on the first two set of motives, as fund-level risk management and diversification have been addressed above.

First, the VC firm-specific motives stem from the needs and opportunities to leverage and compensate for the main resources – partner, funds and deal flow (Bygrave 1987). Syndication compensates for a lack of financial resources, thus offering a means to participate in larger deals than would be feasible alone, it extends the selection and value-adding capabilities and the market-specific knowledge of partners, and it facilitates the acquisition of pre-screened deal flow through syndication invitations from other VC firms (see Table 1 for references). As discussed above, while on the venture level the acquisition of these resources may be seen as stemming from the functional antecedent, on the VC firm level the resource motives are more strategic in nature, should the syndication be used to leverage or compensate for the existing resources.

In contrast, the second category of VC firm-level motives, the management of both the external perceptions of VC firms and interorganizational relationships, is indirect in terms of their contribution to the performance of investment targets and thus can be more easily characterized as stemming from the strategic antecedent. Venture capitalist firms have a documented tendency to window-dress, that is, to seek associations with events, such as successful exits, that enhance the external perceptions of the VC firm (Lerner 1994). Additionally, assessments of a VC firm derived from its associations with central or high-status syndication partners contribute to these external perceptions (Milanov and Shepherd 2008; Podolny 2001). That is, syndication offers a means both to acquire a stake in a late-stage venture approaching an exit and to establish contacts with high-status partners. These enhance the perceptions of the VC firm's performance among LPs and prospective syndication partners, especially if the focal VC firm does not yet have presentable returns from its funds, which is often the case when the second fund is being raised.

Firm- and context-specific contingencies

How relevant are these motivations, and are some of them more pronounced depending on the characteristics of the VC and the context it operates in? On the

basis of existing, mostly pairwise, comparisons of motives, it is hard to draw a coherent picture. In general, Brander *et al.* (2002) conclude that value added is a more significant motive than selection. Bygrave (1987) suggests that deal flow dominates risk reduction, which, according to the results of De Clercq and Dimov (2004), shapes syndication more than knowledge-based motives. Finally, financial motivations are more important than resource-based motivations (Lockett and Wright 1999, 2001).

The inconclusiveness of these results may to some extent be explained by firm- and context-specific factors. When analysing the relative importance of motivations with respect to the investment focus of VC firms, Manigart *et al.* (2006) found a clear hierarchy of motivations for later-stage investors, where financial motives were followed by deal flow, value adding and selection. Although for early-stage investors the value-adding motive was more significant, the results did not reflect a clear hierarchy. Similarly, Lockett and Wright (2001) observed that resource and financial motives were more emphasized for early-stage than later-stage investors. Furthermore, the size of the VC firm seems to increase the importance of deal flow motivation (Verwaal *et al.* 2010), specialization reduces the importance of the selection motive (Manigart *et al.* 2006), and reputation and status affect the syndication negatively and positively, respectively (Dimov and Milanov 2010). To complicate the comparison further, the data in each of the motive-comparison studies are drawn from different contexts.³ While the role of context with respect to motives has not been studied, the existing evidence on the contextual factors suggests that factors such as the legal and institutional environment (Cumming *et al.* 2010), the liquidity of the exit market (Cumming *et al.* 2005) and the intensity of the competition (Lockett and Wright 1999) are likely to affect the use of syndication. Therefore, while the research is inconclusive, it is evident that the interactions between the needs and opportunities of a firm, its specific characteristics and the context in which it operates are likely to result in a fairly complex overall picture regarding what drives syndication for a specific VC in a specific context.

³Brander *et al.* (2002) examine Canadian firms, Bygrave (1987) uses the largest US firms between 1966 and 1992, De Clercq and Dimov (2004) all US firms between 1990 and 2001, Lockett and Wright (1999, 2001) use data from the UK, and finally, the sample of Manigart *et al.* (2006) is based on the markets of six different nations.

Research gaps: decision to syndicate

When assessed against the questions of how, why and when syndication affects performance, the current literature on motivations to syndicate shows at least three gaps.

Gap 1: Motives and their antecedents. First, the research on syndication has been geared towards the identification and mostly pairwise comparison of the motives for syndication. While this is helpful for charting the dimensions of the decision to syndicate, the focus on the effects sought from the syndication has excluded the drivers that enable, force and motivate VCs to syndicate. Therefore, there is a gap in the research when it comes to differentiating the causes of syndication from its effects. Stated in terms of antecedents, this translates into a lack of fine-grained understanding regarding to what extent and when syndication is driven by the functional and strategic pursuits of VC firms.⁴ Adopting a more cause-based approach would allow more theory-driven and context-sensitive approaches to syndication.

Gap 2: Contingency of syndication. This approach would become readily applicable when facing the next two gaps. On the one hand, as suggested by the research, the decision to syndicate appears highly contingent on both the characteristics of the VC firm and contextual factors. Therefore, to address the first gap, we need to understand not only the individual motives, but also the strategic positions of individual VC firms and how these positions affect both the availability of syndication and its effects on the performance of the VC.

Gap 3: Syndication as part of VC's strategy. On the other hand, in order to address the evident contingencies, a VC firm's approach to syndication needs to be seen as a part of its overall strategy and positioning with respect to its resources and investment focus, as well as its social and reputational standing with

⁴As noted above, in some cases, the distinction is more clear-cut, e.g. when syndication is necessitated by agency concerns (functional) or when VCs aim to establish their reputation through seeking association with successful ventures in their later stages (strategic). In other cases, such as syndicating for the purpose of acquiring an additional assessment of the quality of a venture, syndication may be considered functional on the venture level, but strategic if used continuously to compensate for a lack of selection capabilities.

respect to other VC firms. This perspective is not developed enough to provide an answer, thus presenting the third gap.

Syndicate structures

A syndicate is the outcome of a decision to syndicate and, as such, it is the vehicle through which VCs provide the venture with the financial, social and human resources that motivated the VCs to syndicate in the first place. It combines the interests of multiple investors and exists for a number of years before the venture is exited from, and therefore requires co-ordination and interaction between the venture and the VCs. Therefore, one could expect the scope and complexity of issues related to the syndicates to outweigh those related to the decision. However, the structuring, composition and dynamics of syndicates have received only very limited attention, while the selection of syndication partners has drawn significantly more contributions.

The roles of the lead and non-lead investors

It should be noted that the discussion on the decision to syndicate and the research on which it is based are grounded in the perspective of a VC firm faced with a decision as to whether to syndicate an investment or invest in it alone. However, the successful creation of a syndicate requires there to be other VC firms that are willing to participate in the syndicate. While the motives both for forming a syndicate and for joining it are largely similar, joining and, specifically, joining as a non-lead investor, presents at least two operational benefits.

On the one hand, participating in a syndicate as a non-lead investor enables a VC to increase the size of its portfolio with a significantly lower commitment of management effort when compared with non-syndicated investments (Jääskeläinen *et al.* 2006). The management of a syndicated investment is typically the responsibility of a lead investor, who manages both the venture and the syndicate, is more often involved in the board, is more frequently in interaction with the venture, and is more hands-on with the monitoring (Wright and Lockett 2003). Consequently, the non-lead investor spends one-tenth of the time on the management of a syndicated venture that the lead investors do (Gorman and Sahlman 1989), making it possible to leverage the management resources and increase portfolio size.

On the other hand, invitations to syndicate constitute a source of deal flow which extends the flow of proposals coming directly from entrepreneurs. This both extends the opportunity set of the invited VC firm (Bygrave 1987) and reduces the workload required for screening investment proposals, as they come pre-screened by the inviting VC.

Consequently, while this has not been directly addressed by the research, the benefits of syndication to VC firms that join a syndicate permit the reasonable assumption that the creation of a syndicate is not restricted by the availability of syndication partners. However, more significant concerns relate to how VC firms forming syndicates are able to syndicate with the VC firms they prefer, and how VC firms joining syndicates are able to generate opportunities for further syndication. These questions boil down to the issue of how VC firms select their syndication partners.

Partner selection

The literature suggests two competing rationales for the selection of syndication partners. First, starting from the functional antecedent and from the objective of VC firms of generating the highest possible return on their investments, the key factor when selecting partners stems from the perceived contribution of prospective syndication partners to the development and eventual exit value of the investment target. Accordingly, the lead investors' goal is to select those syndication partners for each deal that maximize the value of the specific venture by contributing most to the post-investment management of portfolio ventures through the complementarity of expertise and contacts (Lockett and Wright 2001; Manigart *et al.* 2006; Meuleman *et al.* 2010) and to the exit value of the investment by providing both certification for the value of the venture (Megginson and Weiss 1991) and an association with investment banks and investors (Pollock *et al.* 2004). However, the empirical evidence supporting a preference for the value-adding capabilities of syndication partners is scarce and indirect. While VC firms have been observed to prefer experienced and reputable partners (Hopp 2008; Lerner 1994; Lockett and Wright 1999; Meuleman *et al.* 2009), it is unclear to what extent this is due to a focus on expected contributions or to an attempt by the VC to reduce the uncertainty related to the selection of partners (e.g. see Dimov and Milanov 2010).

The second stream of research, corresponding to the strategic antecedent, is aligned with the general

literature on the selection of alliance partners and focuses on the effects of existing dyadic and network ties among VC firms and the perceptions of the VC firm as a partner. In line with the alliance literature (e.g. Chung *et al.* 2000; Gulati and Gargiulo 1999), the syndication literature suggests that, from a set of potential syndication partners, lead investors prefer to work with VCs: (1) with whom they have a direct or indirect previous relationship, as this increases the perceived level of trust (Meuleman *et al.* 2009; Sorenson and Stuart 2008; Trapido 2007); (2) that are similar to the focal VC with respect to investment focus, experience and success (Du 2009; Trapido 2007); and (3) that have a central or high-status position in the syndication network (Chung *et al.* 2000; Dimov and Milanov 2010; Meuleman *et al.* 2009, 2010). In addition, there is emerging evidence on the social embeddedness of partner selection. On the one hand, the educational and professional backgrounds of VC professionals have a significant impact on which VC firms work together (Rider 2009) and, on the other hand, in this larger context of social interactions, the syndication invitation has been observed to have some characteristics of a gift (Ferrary 2003, 2010).

Irrespective of whether VC firms favour deal-specific contributions or relationship-specific contributions, the motives for partner selection create a link to the decision to syndicate as a VC firm's ability to attract both partners to its out-syndicated deals and invitations to syndicate in are contingent on its characteristics as a syndication partner (Dimov and Milanov 2010; Meuleman *et al.* 2009). Consequently, the compositions of the resulting syndicates are therefore outcomes of the needs and opportunities of the participating VCs.

Composition and dynamics of syndicates

Apart from the general structure of syndicates and partner selection, the research has paid only limited attention to the composition of syndicates and dynamics within them, and only individual studies exist on the topic. While the composition of a syndicate has an empirical association with involvement in ventures (Bottazzi *et al.* 2008; De Clercq *et al.* 2008; Elango *et al.* 1995; Kaplan and Stromberg 2004), governance (Kaplan and Stromberg 2003; Tian 2011), performance (Birmingham *et al.* 2003; Dimov and De Clercq 2006; Giot and Schwenbacher 2007; Guler 2007) and venture internationalization (Mäkelä and Maula 2005), the main interest of most

of these studies has been elsewhere. Detailed aspects of syndicates, such as composition, size and contracting, have received only secondary attention or functioned as an alternative hypothesis to be controlled for.

A single stronger thread of research has focused on the effects of syndicates during the termination of investments. Birmingham *et al.* (2003) and Guler (2007), addressing the escalation of commitment to ventures, that is, the failure to terminate low-quality ventures, observed that the size of a syndicate negatively affected its capability to terminate bad investments. In addition, Kotha (2008) observed that the lack of a syndicate member with a larger stake leads to a decreased likelihood of a successful exit, suggesting that involvement and contributions to the development suffer from free-riding if there is no clear lead investor. Dimov and De Clercq (2006) measured the effect on portfolio level, and found that the larger the average syndicate in which a VC firm is involved, the greater the failure rate of the portfolio.

Research gaps: syndicate structures

In terms of our guiding questions regarding the performance effects of syndication – how, why and when – one could expect that the composition and dynamics of syndicates are central to the issue, and thus much of the research regarding the actual mechanisms of value creation would address syndicates. However, the literature has addressed mainly the specifics of partner selection, therefore presenting significant gaps for further research.

Gap 1: Syndicate composition and dynamics. Despite the centrality of the composition of the syndicate, there is a significant gap and very little research on the topics that could help us to answer the question of through which specific mechanisms syndication affects performance and how the contributions and roles of individual syndicate members are related to these mechanisms. While the decisions to syndicate and the motives based on the suggested effects of syndication imply the existence of such mechanisms that affect the performance of both venture and VC firms, the evidence is indirect. Additionally, while partner selection appears to be based on similar mechanisms, this evidence is equally implied and indirect.

Gap 2: VC's strategy and opportunities to syndicate. The second set of research gaps related to syndicates derives from partner selection. First, if

the issues of partner selection are central to the formation of syndicates, then, as suggested by Dimov and Milanov (2010), the desirability of a VC as a syndication partner affects its opportunities to syndicate. The question of how these opportunities combine with needs, and how they affect the ways in which VCs use syndication as a part of their overall strategy, points towards a gap in the research.

Gap 3: Management of partner portfolio. Additionally, if the partner selection is consequential, and the opportunities with respect to attracting a desirable partner are a relevant question, this implies that, as part of their syndication strategy, VC firms should also manage their partnership portfolios. The questions of whether this is relevant and whether VCs engage in partner portfolio management have not received attention.

Effects of syndication on performance

As it appears that both the decision to syndicate and the composition of syndicates are strongly motivated by their effects on performance, how, then, do these aspects affect the performance of ventures, funds and VC firms? As the majority of the motivations for syndication closely match the mechanisms through which it can affect the value of ventures, it is only to be expected that such a connection exists. Indeed, the extant evidence on the effects of syndication points in the direction of syndication being associated with enhanced performance on the venture level. However, its effects on the performance of funds and VC firms can be considered inconclusive at best. The following summarizes the proposed mechanisms for enhanced performance on the level of ventures, funds and VC firms, and assesses the extant evidence. The effects examined in each of the studies are summarized in Table 2.

Venture-level effects

On the venture level, the research on the motivation and performance effects of syndication suggests two primary mechanisms through which syndication may affect the performance of ventures: the post-investment management and the exit process. First, syndication is likely to enhance the performance of individual investments as a result of the pooling of at least partially unique resources and the contributions of the VC firms participating in a syndicate. Second,

in addition to the resources directly committed to the venture and mediated through the contacts of the VC firms, the existence of an investor group contributes to perceptions of the venture, in terms of both its legitimacy and credibility. On the one hand, the affiliations of a venture with prominent organizations function as symbols of legitimacy and social standing (Higgins and Gulati 2003; Stuart *et al.* 1999), facilitating access to additional opportunities, resources and partners (Stuart *et al.* 1999). On the other hand, the syndicates provide increased reputational capital which helps to certify the quality and correctness of the pricing during the exit process, thus reducing the discount resulting from asymmetric information and enhancing the returns on the investment (Megginson and Weiss 1991).

The evidence from the venture-level examinations of the performance effects demonstrates that syndication has mainly positive effects on the performance of ventures. Studies examining the use of syndication (Brander *et al.* 2002; Das *et al.* 2011), the size of the syndicate (Brander *et al.* 2002; Giot and Schwienbacher 2007; Nahata 2008) and the composition of the syndicate (De Clercq and Dimov 2008) have found them to be generally positively associated with both the returns generated by the investment targets and the time to and probability of a successful exit. However, one should be cautious when weighing the evidence for the benefits. On the one hand, the evidence on performance effects is based largely on the type of exit and the time to exit. While the arguments for value-adding and certification are plausible, the possibility of reversed causality appears equally plausible. That is, successful ventures grow large, thus requiring financing from larger syndicates both to provide sufficient financial resources and to control risks and exposure. On the other hand, the published evidence from investment returns demonstrates contradictory results. Fleming (2004) found that, in Australian investments, syndication was associated with reduced performance, while Brander *et al.* (2002) reported a positive relationship in a Canadian sample. While emerging research (Cumming and Walz 2010; Hege *et al.* 2009) provides supporting evidence for a positive effect on the return on investment, the question regarding to what extent the results are affected by reversed causality remains unaddressed.

VC firm-level effects

As argued above, venture capital funds are the main product offered to the customers of VC management

Table 2. Effects of syndication on performance and outcomes of ventures, funds and firms

Unit of analysis	Syndication measure	Performance measure	Effect	Study
<i>Level: Venture</i>				
Syndication, binary measure		IRR	+	Brander <i>et al.</i> (2002), Cumming <i>et al.</i> (2010)
			–	Fleming (2004)
Syndicate size		Time to exit	–	Das <i>et al.</i> (2011)
		Multiple	+	Das <i>et al.</i> (2011)
		Exit type	+	Das <i>et al.</i> (2011), Tian (forthcoming)*
		IRR	+	Brander <i>et al.</i> (2002),Hege <i>et al.</i> (2009) (in EU)
			None	Hege <i>et al.</i> (2009) (in US)
		Time to exit	–	Giot and Schwienbacher (2007)
		Exit type	(+)	De Clercq and Dimov (2008)
		Exit type	+	Tian (forthcoming)*
		Size and likelihood of exit	+	Nahata (2008)
		Risk of success (IPO/acquisition)	None	Guler (2007)
Syndicate composition	Prior experience with partners	Exit type	+	De Clercq <i>et al.</i> (2008)
	Partner knowledge endowment	Exit type	+ ^a	De Clercq <i>et al.</i> (2008)
	Partner heterogeneity	IPO/acquisition exit	+	Du (2009)*
	Lead with larger share	IPO/acquisition exit	+	Kotha (2008)*
	Type of partner	IRR	None	Mason and Harrison (2002)
Network	Network centrality	Time to exit	+	Hochberg <i>et al.</i> (2007)
		Survival of venture	+	Hochberg <i>et al.</i> (2007)
	Network contacts	IPO	+	Walske <i>et al.</i> (2007)
		Size and likelihood of exit	+	Nahata (2008)
		Risk of exit	+	Jääskeläinen and Maula (2011)*
<i>Level: Fund</i>				
Syndication	Frequency	Profitability index (fund)	+	Gottschalg and Gerasymenko (2008)*
		IRR	None	Gottschalg and Gerasymenko (2008)*, Jääskeläinen <i>et al.</i> (2008)*
Syndicate	Lead with larger share (0/1) Size	Exit ratio	+	Kotha (2008)*
		Profitability index (fund)	None	Gottschalg and Gerasymenko (2008)*
Network	Centrality	Exit ratio	–	Dushnitsky and Shapira (2010)
		Exit rate of a fund	+	Hochberg <i>et al.</i> (2007), Abell and Nisar (2007)
<i>Level: Firm</i>				
Syndication	Use of syndication	Financial performance	+	Hill <i>et al.</i> (2009)
	Syndication frequency	Number of IPOs	None ^b	Jääskeläinen <i>et al.</i> (2006)
	Syndicate size	Failure rate	+	Dimov and De Clercq (2006)
Network	Network embeddedness	Number of successful IPOs	None ^c	Echols and Tsai (2005)
	Status	Survival	+	Bothner <i>et al.</i> (2008b)*
	Status volatility	Growth in number of investment targets	–	Bothner <i>et al.</i> (2008a)*
	Centrality	IPO share	+	Checkley <i>et al.</i> (2010)

*Working paper.

^aEffect moderated by prior experience with partner.^bPositively moderates the effects of portfolio size.^cPositively moderates the effects of niche.

firms, that is, to institutional investors. The VC firm's performance in terms of fund returns then translates into its ability to attract investors to new funds and thus to continue its operations, which effectively is the ultimate measure of a VC firm's performance. Therefore, the questions of interest are how the

venture-level benefits from syndication aggregate to the fund level and how they lead to performance and survival differences among VCs.

In principle, the fund- and firm-level mechanisms enhancing performance are based on the joint effort in selecting and managing investments. This lever-

ages the financial and personnel resources of individual VCs, creating an opportunity for both VC firm-level resource acquisition strategies and fund-level diversification. First, syndication allows VC firms to access the expertise, funds and deal flow of their partners, thus both increasing the amount and variety of resources available and compensating for those skills or connections the VC firm lacks (e.g. Dimov and Milanov 2010; Manigart *et al.* 2006). This can be expected to enhance the quality of the deal flow and the consequent investment decisions, as well as the VC's capabilities and the efficiency with which the portfolio is managed. Second, syndication provides a mechanism to share the workload and reduce the commitment of resources in individual investments, making it possible for VCs to invest in larger targets (Manigart *et al.* 2006) and maintain a larger number of investments (Cumming 2006b; Jääskeläinen *et al.* 2006). This in turn offers a means to lower the risk of the portfolio, both by reducing the exposure in individual deals and by providing for some, although not perfect, diversification (Lockett and Wright 2001; Wilson 1968).

Studies addressing the question of how these two mechanisms contribute to the performance of individual funds are nearly non-existent.⁵ Two unpublished reports examining the effects of syndication on fund returns find that the frequency of syndication, i.e. the ratio of syndicated ventures to all the investments of the fund, appears to have a positive effect on the profitability index but no effect on the internal rate of return (IRR) funds (Gottschalg and Gerasymenko 2008; Jääskeläinen *et al.* 2008). The firm-level performance effects have been assessed through investment outcomes in terms of perceptions of performance (Hill *et al.* 2009), the number and share of successful exits produced (Echols and Tsai 2005; Hochberg *et al.* 2007; Jääskeläinen *et al.* 2006) and the growth and survival of the VC (Bothner *et al.* 2008a, 2008b). These studies imply two concerns for drawing conclusions regarding the effects of syndication on performance. First, as with the venture-level effects, the VC's association with outcomes that can be categorized as successful for the venture does not imply that investments made

in those ventures are successful in terms of returns generated for the VC fund. The measures used in the extant research treat both the entry rounds and the terms of investments as a black box, implying that we simply do not know what VC firms get for their money, and how the returns are divided among investors. Second, as suggested by the studies focusing on the contacts and network positions of VCs (Echols and Tsai 2005; Hochberg *et al.* 2007; Jääskeläinen and Maula 2011), syndicate and partner portfolio composition (De Clercq and Dimov 2008) and the internal factors of a VC, such as portfolio size (Jääskeläinen *et al.* 2006), the benefits of syndication appear to be contingent on the external and internal conditions that VC firms face.

Therefore, the question is about when, with whom and through which strategies VCs can benefit from syndication. That is, as the use of syndication is pervasive, the mere indicator of whether a venture is syndicated or not or whether a VC firm uses syndication or not does not amount to a source of difference between VC firms and funds in terms of performance. It appears that the benefits of syndication are derived from the syndication strategy, not from the use of syndication itself.

Research gaps: syndicate performance

Gap 1: Effect on VC performance. While previous research has paid considerable attention to the motivations for syndication, the evidence regarding whether and how the financial returns of VC funds are affected by syndication has remained inconclusive to some extent. While indirect evidence exists, there is a need for both direct financial evidence on the level of ventures and funds and for more robust indirect measures and methodological approaches that could at least partially remedy the concerns about reversed causality.

Gap 2: Appropriation of the benefits. Second, while the earlier studies are informative on the aspects of syndication behaviour, they do not answer the question of how different VCs might benefit differently from the syndication of their investments. While these studies generally point towards enhanced performance on the venture level, they have assumed considerable homogeneity among VCs regarding syndication. This approach has ignored the effects of syndicate roles and the round of entry on how VCs benefit from individual investments. In short, most of the studies that rely on the performance of target

⁵Venture capitalists have been very secretive regarding the actual returns on investments, and the data on returns have only recently become available for researchers. However, depending on the source, the data are far from exhaustive (e.g. private equity intelligence) or only selectively available to researchers (e.g. venture economics).

companies fail to consider what price the investors paid when investing and what price they receive when exiting the investment. The research on performance effects has not yet converged with the approaches of research addressing the contingencies affecting syndication behaviour (e.g. Bygrave 1987; Lerner 1994; Lockett and Wright 2001; Manigart *et al.* 2006).

Directions for further research

Current gaps in understanding of syndication

It appears that what is known and what we need to know can be roughly divided with respect to the two antecedents of syndication. From a functional perspective, the existing research has focused strongly on decisions as to when to syndicate and with whom to syndicate. Focusing on the needs of ventures, the research has identified, compared and validated multiple motives for syndication. Additionally, the venture-level effects of syndication and syndicates provide strong, although not unproblematic, evidence for the benefits of syndication.

The gaps that were identified within each of the three segments of the literature point almost exclusively towards the role of VC-level factors and the strategic antecedent. First, with respect to the decision to syndicate, there is a lack of research that would disentangle causes and effects regarding the motives to syndicate. We consider this to be due to both the lack of understanding of the firm- and context-specific contingencies and the underdeveloped strategic approach towards syndication, which has resulted in a venture-specific focus, with the strategic concerns of VCs being neglected. Second, in relation to syndicates, there is a lack of research regarding the actual mechanisms through which syndication affects performance. Taken together with the gap related to the relative importance of firm- and venture-level motivations, we have difficulties in assessing whether it matters who VCs include in syndicates as co-investors. Consequently, this also makes it hard to assess whether the gap regarding the management of relationships and partner portfolios is a significant one, as we do not know the relative importance of strategic factors and the effects of syndicate partners on performance. Finally, reflecting the above gaps, we do not know whether syndication affects firm-level performance, whether it should and, if it should, then when.

Current gaps in approaches to syndication

The lack of research with a strategic orientation is evident when analysing the streams of literature and theoretical approaches that the existing studies have drawn on. As presented in Table 3, current approaches are strongly characterized by four main categories: (1) syndication-driven articles drawing from VC-level theorizing; (2) mostly formal approaches stemming from the financial tradition; (3) studies of networks and interorganizational relationships where syndication acts as an empirical context; and (4) a collection of disperse contributions from other perspectives that either use syndication as a context or seek to offer more theoretical arguments about individual aspects of syndication. The syndication literature has been highly phenomena-driven, offering only limited theoretical contributions other than the formal approaches from the financial literature. Therefore, there is an evident need to broaden the theoretical approaches to syndication. Especially revealing is that of the more than 70 studies, only four draw directly on either resource-based theory or transaction cost economics. Given the central roles of resource concerns and co-operation in syndication, one could expect these perspectives to enrich the current approaches.

In addition to the mainly VC-level theoretical approaches, the methodological and empirical basis of the research presents equal uniformity, as shown in Table 4. Large investment databases, primarily the VentureXpert (VE) of ThomsonReuters, have offered an accessible source of data documenting syndicated investments, syndicate compositions and the outcomes of syndicated ventures. While they do offer information on large samples of firms, the richness of secondary data sources is naturally limited. In terms of primary data, there have been a number of survey-based contributions which have been instrumental in broadening the perspectives on syndication, especially on motives, contingencies and compositions (e.g. Lockett and Wright 1999, 2001; Manigart *et al.* 2006; Wright and Lockett 2003). However, given the gaps in the research regarding the dynamics of syndicates and contingencies of syndication, there is a need for a more fine-grained understanding from methodological approaches that are able to tap into the processes and practices of VCs and thereby derive insights that are missing from the current approaches. Such approaches have been rare in syndication research, thus presenting a promising area for further research.

Table 3. Theoretical approaches to syndication

Main theoretical lens	Decision and motivation	Composition and dynamics	Effects on performance
VC-level theorizing	Bygrave (1987, 1988), Cumming (2005), Cumming <i>et al.</i> (2010), Dal-Pont Legrand and Pommet (2010), Deli and Santhanakrishnan (2010), Fritsch and Schilder (2008), Lerner (1994), Lockett and Wright (1999, 2001), Manigart <i>et al.</i> (2006), Michie (1981)	Bottazzi <i>et al.</i> (2008), Checkley and Steglich (2007), Cumming <i>et al.</i> (2005), Dimov and De Clercq (2006), Mäkelä and Maula (2005, 2006, 2008)	Brander <i>et al.</i> (2002) Checkley (2009), Cumming (2006b), Cumming and Dai (2010), Cumming <i>et al.</i> (2010), Das <i>et al.</i> (2011), Filatotchev <i>et al.</i> (2005, 2006), Fleming (2004), Dimov and De Clercq (2006), Giot and Schwenbacher (2007), Hill <i>et al.</i> (2009), Jääskeläinen <i>et al.</i> (2006), Kaiser and Lauterbach (2007), Mason and Harrison (2002)
Contracting, agency, asymmetric information	Admati and Pfleiderer (1994), Biais and Perotti (2008), Casamatta and Haritchabalet (2007), Cumming (2006a), Huang and Xu (2003), Meuleman <i>et al.</i> (2009), Schmidt (2003), Tian (2011)	Hellmann (2002), Kaplan and Stromberg (2003, 2004), Meuleman <i>et al.</i> (2010), Tykvová (2007)	Cumming and Dai (2010), Dushnitsky and Shapira (2010), Nahata (2008)
Networks and interorganizational relationships	Dimov and Milanov (2010)	De Clercq <i>et al.</i> (2008), Hopp (2008, 2010a, 2010b); Hopp and Rieder (2011), Keil <i>et al.</i> (2010), Meuleman <i>et al.</i> (2010), Sorenson and Stuart (2008, 2001), Trapido (2007), Wright and Lockett (2003), Mote (2011)	Abell and Nisar (2007), Checkley <i>et al.</i> (2010), Dimov <i>et al.</i> (2007), Echols and Tsai (2005), Fund <i>et al.</i> (2008), Hochberg <i>et al.</i> (2007), Hochberg <i>et al.</i> (2010), Kogut <i>et al.</i> (2007), Podolny (2001), Walker (2008), Walske <i>et al.</i> (2007)
Other	<i>RBV/KBV</i> De Clercq and Dimov (2004), Ferrary (2010), Verwaal <i>et al.</i> (2010) <i>Learning</i> Tian (2011), Meuleman and Wright (2011) <i>TCE</i> Verwaal <i>et al.</i> (2010)	<i>Gift exchange</i> Ferrary (2003, 2010)	<i>RBV/KBV</i> De Clercq <i>et al.</i> (2008) <i>Decision-making</i> Birmingham <i>et al.</i> (2003), Guler (2007)

These shortcomings in understanding and approaches suggest two primary directions for further research. First, reflecting the lack of strategic and contextual understanding, we suggest that research would benefit from addressing syndication as part of its overall strategy, and especially considering the role of resources. Second, more research is needed with respect to the performance effects, both in terms of VC performance and survival and in terms of how the aspects of syndication combine on the level of the industry and how it affects the LPs.

Direction 1: Syndication, VC firm strategy and resources

Recent literature focusing on the strategic dimensions of venture capital firms has begun to investigate

the connections between VCs' resource endowments and their actions and performance. In this sphere, studies have reported that VCs select their niche position (Echols and Tsai 2005) and investment focus (Dimov and De Clercq 2006), as well as choosing their level of involvement in portfolio companies (De Clercq *et al.* 2008; Gifford 1997; Jääskeläinen *et al.* 2006), on the basis of their endowments in human and social resources. Given the role of syndication in leveraging and compensating VC firms' resources for value adding, selection and deal flow, it can be expected that the extent to which a VC firm uses syndication will be connected to its resource endowment. Indeed, in a recent study, Verwaal *et al.* (2010) observed that the resource needs and access capabilities of VCs mediate the relationship between firm size and syndication frequency. To address these resource contingencies of syndication strategy, we

Table 4. Methodological approaches and data sources

Theory	Decision and motivation	Composition and dynamics	Effects on performance
Secondary data	<i>VE</i> : Bygrave (1987, 1988), De Clercq and Dimov (2004), Dimov and Milanov (2010), Deli and Santhanakrishnan (2010), Lerner (1994), Tian (2011) <i>CEPRES</i> : Cumming <i>et al.</i> (2010) <i>CMBOR</i> : Meuleman <i>et al.</i> (2009) Meuleman and Wright (2011) <i>MacDonald & Associates</i> : Cumming (2005), Cumming (2006a) <i>VC Facts</i> : Fritsch and Schilder (2008) <i>PWC</i> : Ferrary (2010)	<i>VE</i> : De Clercq <i>et al.</i> (2008) Cumming <i>et al.</i> (2005), Hopp (2008), Keil <i>et al.</i> (2010), Mote (2011), Sorenson and Stuart (2008, 2001), Trapido (2007) <i>EI Consulting</i> : Checkley and Steglich (2007) <i>CMBOR</i> : Meuleman <i>et al.</i> (2010) <i>PWC</i> : Ferrary (2010)	<i>VE</i> : Abell and Nisar (2007), Birmingham <i>et al.</i> (2003), Cumming <i>et al.</i> (2010), De Clercq <i>et al.</i> (2008), Dimov and De Clercq (2006), Das <i>et al.</i> (2011), Dimov <i>et al.</i> (2007), Dushnitsky and Shapira (2010), Echols and Tsai (2005), Giot and Schwienbacher (2007), Guler (2007), Hochberg <i>et al.</i> (2007), Hochberg <i>et al.</i> (2010), Jääskeläinen <i>et al.</i> (2006), Kogut <i>et al.</i> (2007), Nahata (2008), Podolny (2001), Walker (2008) <i>CEPRES</i> : Cumming and Dai (2010), Kaiser and Lauterbach (2007), <i>MacDonald & Associates</i> : Brander <i>et al.</i> (2002) Cumming (2006b), <i>EI Consulting</i> : Checkley (2009); Checkley <i>et al.</i> (2010)
Survey	Lockett and Wright (1999, 2001), Manigart <i>et al.</i> (2006), Verwaal <i>et al.</i> (2010)	Bottazzi <i>et al.</i> (2008), De Clercq <i>et al.</i> (2008), Wright and Lockett (2003)	Fleming (2004), Hill <i>et al.</i> (2009), Mason and Harrison (2002)
Other primary quantitative data		Kaplan and Stromberg (2003, 2004)	Filatotchev <i>et al.</i> (2005, 2006)
Qualitative approaches	<i>Case</i> : Michie (1981)	<i>Case</i> : Ferrary (2003) <i>Grounded theory</i> : Mäkelä and Maula (2005, 2006, 2008)	<i>Case</i> : Fund <i>et al.</i> (2008) <i>Grounded theory</i> : Walske <i>et al.</i> (2007)
Formal models	Admati and Pfleiderer (1994), Biais and Perotti (2008), Casamatta and Haritchabalet (2007), Dal-Pont Legrand and Pomet (2010), Huang and Xu (2003), Schmidt (2003)	Hellmann (2002), Tykvová (2007)	

suggest three questions for further research which are based on the identified gaps.

How do the resource strategies of VC firms affect syndication? While co-operation with other VCs should affect the efficiency and effectiveness of a VC firm positively by leveraging its existing resources, the use of these external resources and reliance on them also makes an organization dependent on the continuity and recurrence of the co-operation (Pfeffer and Salancik 1978). Depending on the market, on average 30–40% of the investments of VCs are made following syndication invitations from other VCs (Jääskeläinen *et al.* 2006; Manigart *et al.* 2006). Whether the extensive use of syndication increases the risks of a VC firm or affects its performance negatively is an interesting topic for further research.

How do VCs manage their portfolio of syndication partners? In addition to the direct effects of resource leverage, the benefits of syndication and the mechanisms of partner selection suggest that finding new syndication partners and managing relationships with existing ones are relevant concerns for VC firms. The literature on alliances has recently directed increasing attention to the management of partner portfolios as part of a firm's overall strategy (Hoffmann 2007). For example, issues such as the exploration and exploitation of syndication partners (Lavie and Rosenkopf 2006), the diversity of partners (Goerzen and Beamish 2005) and the effects of repeated co-operation (Goerzen 2007) also appear directly relevant within the context of VC firms. In essence, the key question here is how the leveraging of VC firms' resources through syndication affects the management of these interorganizational

relationships that provide access to additional resources in the first place.

How do the characteristics of a VC as a syndication partner affect syndication strategy? Furthermore, the opportunities for resource leverage and the need to manage the partner portfolio are likely to be contingent on the characteristics of the VC firm as a syndication partner. The consideration of reputation and status draws attention to the effects of a VC firm's structural standing on its opportunities to pursue specific syndication strategies. A VC firm's ability to leverage operations with the resources of its partners, i.e. with network resources (Lavie and Rosenkopf 2006), is contingent on its ability to attract invitations to syndicate, that is, to syndicate in. Whether these and other strategic concerns dominate venture-specific motives with respect to decisions on whether to syndicate and with whom is a subject for further research.

Direction 2: Syndication and performance

How do the composition and dynamics of syndicates affect the performance of ventures? The fact that the extant research presents no consistent evidence for the fund-level performance effects of syndication presents two opportunities for further research. First, further research is required to either confirm or refute the existing results based on emerging research that, in fact, there is no statistical relationship between syndication and performance. This requires development in terms of identifying the mechanisms through which syndication might affect performance. On the venture level, this implies the examination of the marginal contribution of syndicate members to the performance of ventures and the resource complementarity of syndicate members. While complementary assets have been found to form one of the dominant drivers for the motivations and benefits of interorganizational relationships in general (e.g. Eisenhardt and Schoonhoven 1996; Stuart 2000), the syndication literature has directed only limited attention to the resource complementarity of syndicate partners. So far, aspects such as experience and specialized expertise have received modest attention (e.g. De Clercq and Dimov 2004; Du 2009), but the interplay of the complementary resource of syndicate partners lacks contributions. Therefore, more research is needed in terms of what different types of investors bring to syndicates, how these complementary assets interact, and what their effects on investments are.

Who appropriates the venture-level benefits of syndication? Given that the current evidence comes mainly from US venture capital markets, where approximately 80% of investments are syndicated, it is not surprising that whether syndication is used or not is not a sufficient factor to create performance differences among funds. Thus, it is more interesting to ask whether some VC firms are better positioned to reap the venture-level performance benefits of syndication (Lavie 2007) and how syndication affects the benefits from venture capital in general. On the one hand, studies addressing the effects of VC firms' network positions have found that centrally positioned VC firms may enjoy structural benefits that contribute to their performance (Bothner *et al.* 2008b; Hochberg *et al.* 2007). These benefits include both opportunities to attract better deal flow and opportunities to structure and negotiate the investments in a manner that affects the appropriation of financial returns. On the other hand, system-level studies have found that tight syndication networks both reduce competition and overall diversification (Checkley 2009; Hochberg *et al.* 2010), thereby introducing adverse effects stemming from the strategic actions of VCs. The magnitude of these effects with respect to the overall benefits of syndication is unknown. Therefore, whether some firms benefit more from syndication than others and how much syndication enhances the overall returns from venture capital are interesting open questions for further research.

Summary

In this paper, we set out to review and synthesize the extant research on venture capital syndication. The shape of the review has been influenced by our view of syndicates as a form of horizontal interorganizational co-operation organized around the provision of financing for firms with a financial deficit. While the function of syndication is carved out by VCs' role as financial intermediaries, the use and outcomes of syndication are affected by the strategic concerns of VCs striving for performance and survival. From these functional and strategic perspectives, we sought the answer to the question of how well we understand why VCs syndicate their investments, how the syndicates are structured and how syndication affects performance.

The identification of the two antecedents and the structuring of the current research according to the schematic structure of syndication enabled us to

examine how current contributions are balanced and where gaps in the extant research exist. We found that the research has emphasized the functional antecedent and mainly addressed the motives for syndication. Accordingly, the strategic aspects of syndication are less well understood. We found the research on syndicate dynamics especially lacking, as syndicates are the main vehicle of interaction and value adding, through which the contributions of VCs flow to ventures.

These gaps in the extant research led us to propose two main research themes that summarize the gaps observed in the review of the research. On the one hand, we found that, while conventional wisdom on the benefits of syndication as a means to enhance the performance of the targets has received attention and support from the research, the implications of this performance enhancement for VC firms are based on evidence that is both indirect and scant. The main avenues for further contributions with respect to performance implications are based both on methodological approaches that could counter the evident but non-addressed endogeneity concerns and on the theoretical concerns about bargaining and the appropriation of rents from syndicates. On the other hand, the appropriation of rents implies that one source of performance difference is rooted in the VC firm's ability to form and extract rents from syndicates. Consequently, we found that research on syndication could be greatly improved by viewing syndication as part of VC firms' overall strategy. This approach could potentially lead future research to address the questions of how the differing contexts and resource endowments of VC condition the use of syndication and lead to differing opportunities to benefit from syndication. This strategic approach would deepen our understanding of the principles on which the interorganizational relationships are built in venture capital and, consequently, facilitate the increasing use of VC syndication as an empirical context.

References

- Abell, P. and Nisar, T.M. (2007). Performance effects of venture capital firm networks. *Management Decision*, **45**, pp. 923–936.
- Admati, A.R. and Pfleiderer, P. (1994). Robust financial contracting and the role of venture capitalists. *Journal of Finance*, **49**, pp. 371–402.
- Amit, R., Brander, J. and Zott, C. (1998). Why do venture capital firms exist? Theory and Canadian evidence. *Journal of Business Venturing*, **13**, pp. 441–466.
- Bachmann, R. and Schindele, I. (2006). Theft and syndication in venture capital finance. Available at SSRN: <http://ssrn.com/abstract=896025>.
- Biais, B. and Perotti, E. (2008). Entrepreneurs and new ideas. *Rand Journal of Economics*, **39**, pp. 1105–1125.
- Birmingham, C., Busenitz, L.W. and Arthurs, J.D. (2003). The escalation of commitment by venture capitalists in reinvestment decisions. *Venture Capital*, **5**, pp. 217–230.
- Black, B.S. and Gilson, R.J. (1998). Venture capital and the structure of capital markets: banks versus stock markets. *Journal of Financial Economics*, **47**, pp. 243–277.
- Bothner, M., Kang, J.-H. and Lee, W. (2008a). Status volatility and organizational growth in the U.S. venture capital industry. University of Chicago.
- Bothner, M., Kim, Y.-K. and Lee, W. (2008b). Primary status, complementary status, and capital acquisition in the U.S. venture capital industry. Working paper, University of Chicago.
- Bottazzi, L., Da Rin, M. and Hellmann, T. (2008). Who are the active investors? Evidence from venture capital. *Journal of Financial Economics*, **89**, pp. 488–512.
- Brander, J.A., Amit, R. and Antweiler, W. (2002). Venture-capital syndication: improved venture selection vs. the value-added hypothesis. *Journal of Economics & Management Strategy*, **11**, pp. 423–452.
- Bygrave, W.D. (1987). Syndicated investments by venture capital firms – a networking perspective. *Journal of Business Venturing*, **2**, pp. 139–154.
- Bygrave, W.D. (1988). The structure of the investment networks of venture capital firms. *Journal of Business Venturing*, **3**, pp. 137–157.
- Casamatta, C. and Haritchabalet, C. (2007). Experience, screening and syndication in venture capital investments. *Journal of Financial Intermediation*, **16**, pp. 368–398.
- Castilla, E.J. (2003). Networks of venture capital firms in Silicon Valley. *International Journal of Technology Management*, **25**, pp. 113–135.
- Cestone, G., Lerner, J. and White, L. (2007). The design of syndicates in venture capital. Working paper, CSEF University of Salerno.
- Chan, Y.S. (1983). On the positive role of financial intermediation in allocation of venture capital in a market with imperfect information. *Journal of Finance*, **38**, pp. 1543–1568.
- Checkley, M. (2009). Inadvertent systemic risk in financial networks: venture capital and institutional funds. *Long Range Planning*, **42**, pp. 341–358.
- Checkley, M. and Steglich, C. (2007). Partners in power: job mobility and dynamic deal-making. *European Management Review*, **4**, pp. 161–171.
- Checkley, M., Higón, D.A. and Angwin, D. (2010). Venture capital syndication and its causal relationship with performance outcomes. *Strategic Change*, **19**, pp. 195–212.
- Chung, S., Singh, H. and Lee, K. (2000). Complementarity, status similarity and social capital as drivers of alliance formation. *Strategic Management Journal*, **21**, pp. 1–22.

- Cumming, D. (2006a). Adverse selection and capital structure: evidence from venture capital. *Entrepreneurship Theory and Practice*, **30**, pp. 155–183.
- Cumming, D. and Dai, N. (2010). Local bias in venture capital investments. *Journal of Empirical Finance*, **17**, pp. 362–380.
- Cumming, D. and Walz, U. (2010). Private equity returns and disclosure around the world. *Journal of International Business Studies*, **41**, pp. 727–754.
- Cumming, D., Fleming, G. and Schwienbacher, A. (2005). Liquidity risk and venture capital finance. *Financial Management*, **34**, pp. 77–105.
- Cumming, D., Schmidt, D. and Walz, U. (2010). Legality and venture capital governance around the world. *Journal of Business Venturing*, **25**, pp. 54–72.
- Cumming, D.J. (2005). Agency costs, institutions, learning, and taxation in venture capital contracting. *Journal of Business Venturing*, **20**, pp. 573–622.
- Cumming, D.J. (2006b). The determinants of venture capital portfolio size: empirical evidence. *Journal of Business*, **79**, pp. 1083–1126.
- Dal-Pont Legrand, M. and Pomet, S. (2010). Venture capital syndication and the financing of innovation: financial versus expertise motives. *Economics Letters*, **106**, pp. 75–77.
- Das, S.R., Jo, H. and Kim, Y. (2011). Polishing diamonds in the rough: the sources of syndicated venture performance. *Journal of Financial Intermediation*, **20**, pp. 199–230.
- De Clercq, D. and Dimov, D. (2004). Explaining venture capital firms' syndication behaviour: a longitudinal study. *Venture Capital*, **6**, pp. 243–256.
- De Clercq, D. and Dimov, D. (2008). Internal knowledge development and external knowledge access in venture capital investment performance. *Journal of Management Studies*, **45**, pp. 585–612.
- De Clercq, D. and Dimov, D. (2010). Doing it not alone: antecedents, dynamics and outcomes of venture capital syndication. In Cumming, D.J. (ed.), *Venture Capital: Investment Strategies, Structures, and Policies*. New York: Wiley, pp. 221–242.
- De Clercq, D., Sapienza, H.J. and Zaheer, A. (2008). Firm and group influences on venture capital firms' involvement in new ventures. *Journal of Management Studies*, **45**, pp. 1169–1194.
- Deli, D.N. and Santhanakrishnan, M. (2010). Syndication in venture capital financing. *Financial Review*, **45**, pp. 557–578.
- Dessi, R. (2005). Start-up finance, monitoring, and collusion. *Rand Journal of Economics*, **36**, pp. 255–274.
- Diamond, D.W. (1984). Financial intermediation and delegated monitoring. *Review of Economic Studies*, **51**, pp. 393–414.
- Dimov, D. and De Clercq, D. (2006). Venture capital investment strategy and portfolio failure rate: a longitudinal study. *Entrepreneurship Theory and Practice*, **30**, pp. 207–223.
- Dimov, D. and Milanov, H. (2010). The interplay of need and opportunity in venture capital investment syndication. *Journal of Business Venturing*, **25**, pp. 331–348.
- Dimov, D., Shepherd, D.A. and Sutcliffe, K.M. (2007). Requisite expertise, firm reputation, and status in venture capital investment allocation decisions. *Journal of Business Venturing*, **22**, pp. 481–502.
- Du, Q. (2009). Birds of a feather or celebrating differences? The formation and impact of venture capital syndication: Available at SSRN <http://ssrn.com/abstract=1360545>.
- Dushnitsky, G. and Shapira, Z. (2010). Entrepreneurial finance meets organizational reality: comparing investment practices and performance of corporate and independent venture capitalists. *Strategic Management Journal*, **31**, pp. 990–1017.
- Echols, A. and Tsai, W. (2005). Niche and performance: the moderating role of network embeddedness. *Strategic Management Journal*, **26**, pp. 219–238.
- Eisenhardt, K.M. and Schoonhoven, C.B. (1996). Resource-based view of strategic alliance formation: strategic and social effects in entrepreneurial firms. *Organization Science*, **7**, pp. 136–150.
- Elango, B., Fried, V.H., Hisrich, R.D. and Polonchek, A. (1995). How venture capital firms differ. *Journal of Business Venturing*, **10**, pp. 157–179.
- Ferrary, M. (2003). The gift exchange in the social networks of Silicon Valley. *California Management Review*, **45**, pp. 120–138.
- Ferrary, M. (2010). Syndication of venture capital investment: the art of resource pooling. *Entrepreneurship Theory and Practice*, **34**, pp. 885–907.
- Filatovchev, I., Chahine, S., Wright, M. and Arberk, M. (2005). Founders' characteristics, venture capital syndication and governance in entrepreneurial IPOs. *International Entrepreneurship and Management Journal*, **1**, pp. 419–439.
- Filatovchev, I., Wright, M. and Arberk, M. (2006). Venture capitalists, syndication and governance in initial public offerings. *Small Business Economics*, **26**, pp. 337–350.
- Fleming, G. (2004). Venture capital returns in Australia. *Venture Capital*, **6**, pp. 23–45.
- Fluck, Z., Garrison, K. and Myers, S. (2009). Holding hands or tying hands? A theory of syndication in venture capital investments. MIT Sloan Working Paper.
- Fritsch, M. and Schilder, D. (2008). Does venture capital investment really require spatial proximity? An empirical investigation. *Environment and Planning A*, **40**, pp. 2114–2131.
- Fund, B.R., Pollock, T.G., Baker, T. and Wowak, A.J. (2008). Who's the new kid? The process of developing centrality in venture capitalist deal networks. *Network Strategy*, **25**, pp. 563–593.
- Gifford, S. (1997). Limited attention and the role of the venture capitalist. *Journal of Business Venturing*, **12**, pp. 459–482.

- Giot, P. and Schwienbacher, A. (2007). IPOs, trade sales and liquidations: modelling venture capital exits using survival analysis. *Journal of Banking & Finance*, **31**, pp. 679–702.
- Goerzen, A. (2007). Alliance networks and firm performance: the impact of repeated partnerships. *Strategic Management Journal*, **28**, pp. 487–509.
- Goerzen, A. and Beamish, P.W. (2005). The effect of alliance network diversity on multinational enterprise performance. *Strategic Management Journal*, **26**, pp. 333–354.
- Gompers, P. and Lerner, J. (1999). An analysis of compensation in the US venture capital partnership. *Journal of Financial Economics*, **51**, pp. 3–44.
- Gompers, P.A. (1995). Optimal investment, monitoring, and the staging of venture capital. *Journal of Finance*, **50**, pp. 1461–1489.
- Gorman, M. and Sahlman, W.A. (1989). What do venture capitalists do? *Journal of Business Venturing*, **4**, pp. 231–248.
- Gottschalg, O. and Gerasymenko, V. (2008). Antecedents and consequences of venture capital syndication. Paper presented at the Academy of Management Annual Meeting, Anaheim, CA, August. Available at SSRN: <http://ssrn.com/abstract=1095033>.
- Gulati, R. and Gargiulo, M. (1999). Where do inter-organizational networks come from? *American Journal of Sociology*, **104**, pp. 1439–1493.
- Guler, I. (2007). Throwing good money after bad? Political and institutional influences on sequential decision making in the venture capital industry. *Administrative Science Quarterly*, **52**, pp. 248–285.
- Gupta, A.K. and Sapienza, H.J. (1992). Determinants of venture capital firms preferences regarding the industry diversity and geographic scope of their investments. *Journal of Business Venturing*, **7**, pp. 347–362.
- Hege, U., Palomino, F. and Schwienbacher, A. (2009). Venture capital performance: the disparity between Europe and the United States. *Finance*, **30**, pp. 7–50.
- Hellmann, T. (2002). A theory of strategic venture investing. *Journal of Financial Economics*, **64**, pp. 285–314.
- Higgins, M.C. and Gulati, R. (2003). Getting off to a good start: the effects of upper echelon affiliations on underwriter prestige. *Organization Science*, **14**, pp. 244–263.
- Hill, S.A., Maula, M., Birkinshaw, J. and Murray, G. (2009). Transferability of the venture capital model to the corporate context: implications for the performance of corporate venture units. *Strategic Entrepreneurship Journal*, **3**, pp. 3–27.
- Hochberg, Y., Ljungqvist, A. and Lu, Y. (2010). Networking as a barrier to entry and the competitive supply of venture capital. *Journal of Finance*, **65**, pp. 829–859.
- Hochberg, Y.V., Ljungqvist, A. and Lu, Y. (2007). Whom you know matters: venture capital networks and investment performance. *Journal of Finance*, **62**, pp. 251–301.
- Hoffmann, W.H. (2007). Strategies for managing a portfolio of alliances. *Strategic Management Journal*, **28**, pp. 827–856.
- Hopp, C. (2008). Are firms reluctant to engage in inter-organizational exchange relationships with competitors? *Economics Letters*, **100**, pp. 348–350.
- Hopp, C. (2010a). The evolution of inter-organizational networks in venture capital financing. *Applied Financial Economics*, **20**, pp. 1725–1739.
- Hopp, C. (2010b). When do venture capitalists collaborate? Evidence on the driving forces of venture capital syndication. *Small Business Economics*, **35**, pp. 417–443.
- Hopp, C. and Rieder, F. (2011). What drives venture capital syndication? *Applied Economics*, **43**, pp. 3089–3102.
- Huang, H.Z. and Xu, C.G. (2003). Financial syndication and R&D. *Economics Letters*, **80**, pp. 141–146.
- Jääskeläinen, M. and Maula, M. (2011). Do networks of financial intermediaries help reduce local bias? Evidence from cross-border venture capital. Working paper, Aalto University.
- Jääskeläinen, M., Maula, M. and Pynnä, J. (2008). Syndication strategies and financial performance in venture capital: the moderating effect of investor status. Working paper, Helsinki University of Technology.
- Jääskeläinen, M., Maula, M. and Seppä, T. (2006). Allocation of attention to portfolio companies and the performance of venture capital firms. *Entrepreneurship Theory and Practice*, **30**, pp. 185–206.
- Jääskeläinen, M., Maula, M. and Pynnä, J. (2008). Syndication strategies and financial performance in venture capital: the moderating effect of investor status. Helsinki University of Technology.
- Kaiser, D.G. and Lauterbach, R. (2007). The need for diversification and its impact on the syndication probability of venture capital investments. *Journal of Alternative Investments*, **10**, pp. 62–79.
- Kaplan, S.N. and Stromberg, P. (2003). Financial contracting theory meets the real world: an empirical analysis of venture capital contracts. *Review of Economic Studies*, **70**, pp. 281–315.
- Kaplan, S.N. and Stromberg, P. (2004). Characteristics, contracts, and actions: evidence from venture capitalist analyses. *Journal of Finance*, **59**, pp. 2177–2210.
- Keil, T., Maula, M.V.J. and Wilson, C. (2010). Unique resources of corporate venture capitalists as a key to entry into rigid venture capital syndication networks. *Entrepreneurship Theory and Practice*, **34**, pp. 83–103.
- Kogut, B., Urso, P. and Walker, G. (2007). Emergent properties of a new financial market: American venture capital syndication, 1960–2005. *Management Science*, **53**, pp. 1181–1198.
- Kotha, R. (2008). Equity traps: the distribution of cash flow incentives among investors in venture capital syndicates and performance of start-ups. Available at SSRN: <http://ssrn.com/abstract=1115050>.

- Lavie, D. (2007). Alliance portfolios and firm performance: a study of value creation and appropriation in the US software industry. *Strategic Management Journal*, **28**, pp. 1187–1212.
- Lavie, D. and Rosenkopf, L. (2006). Balancing exploration and exploitation in alliance formation. *Academy of Management Journal*, **49**, pp. 797–818.
- Leland, H.E. and Pyle, D.H. (1977). Informational asymmetries, financial structure, and financial intermediation. *Journal of Finance*, **32**, pp. 371–387.
- Lerner, J. (1994). The syndication of venture capital investments. *Financial Management*, **23**, pp. 16–27.
- Lerner, J. (1995). Venture capitalists and the oversight of private firms. *Journal of Finance*, **50**, pp. 301–318.
- Lockett, A. and Wright, M. (1999). The syndication of private equity: evidence from the UK. *Venture Capital*, **1**, pp. 303–324.
- Lockett, A. and Wright, M. (2001). The syndication of venture capital investments. *Omega – International Journal of Management Science*, **29**, pp. 375–390.
- Mäkelä, M.M. and Maula, M.V.J. (2005). Cross-border venture capital and new venture internationalization: an isomorphism perspective. *Venture Capital*, **7**, pp. 227–257.
- Mäkelä, M.M. and Maula, M.V.J. (2006). Interorganizational commitment in syndicated cross-border venture capital investments. *Entrepreneurship Theory and Practice*, **30**, pp. 273–298.
- Mäkelä, M.M. and Maula, M.V.J. (2008). Attracting cross-border venture capital: the role of a local investor. *Entrepreneurship and Regional Development*, **20**, pp. 237–257.
- Manigart, S. et al. (2006). Venture capitalists' decision to syndicate. *Entrepreneurship Theory and Practice*, **30**, pp. 131–153.
- Mason, C.M. and Harrison, R.T. (2002). Is it worth it? The rates of return from informal venture capital investments. *Journal of Business Venturing*, **17**, pp. 211–236.
- Meggison, W.L. and Weiss, K.A. (1991). Venture capitalist certification in initial public offerings. *Journal of Finance*, **46**, pp. 879–903.
- Meuleman, M. and Wright, M. (2011). Cross-border private equity syndication: institutional context and learning. *Journal of Business Venturing*, **26**, pp. 35–48.
- Meuleman, M., Lockett, A., Manigart, S. and Wright, M. (2010). Partner selection decisions in interfirm collaborations: the paradox of relational embeddedness. *Journal of Management Studies*, **47**, pp. 995–1019.
- Meuleman, M., Wright, M., Manigart, S. and Lockett, A. (2009). Private equity syndication: agency costs, reputation and collaboration. *Journal of Business Finance & Accounting*, **36**, pp. 616–644.
- Michie, R.C. (1981). Options, concessions, syndicates, and the provision of venture capital, 1880–1913. *Business History*, **23**, pp. 147–164.
- Milanov, H. and Shepherd, D. (2008). 'One is known by the company one keeps': imprinting effects of a firm's network entry on its future status. *Frontiers of Entrepreneurship Research*, **28**(15), paper 2.
- Mote, J. (2011). Syndication, networks and the growth of venture capital in Philadelphia, 1980–99. *Industry and Innovation*, **18**, pp. 131–150.
- Nahata, R. (2008). Venture capital reputation and investment performance. *Journal of Financial Economics*, **90**, pp. 127–151.
- Norton, E. and Tenenbaum, B.H. (1993). Specialization versus diversification as a venture capital-investment strategy. *Journal of Business Venturing*, **8**, pp. 431–442.
- Pfeffer, J. and Salancik, G.R. (1978). *The External Control of Organizations: A Resource Dependence Perspective*. New York: Harper & Row.
- Pichler, P. and Wilhelm, W. (2001). A theory of the syndicate: form follows function. *Journal of Finance*, **56**, pp. 2237–2264.
- Podolny, J.M. (2001). Networks as the pipes and prisms of the market. *American Journal of Sociology*, **107**, pp. 33–60.
- Pollock, T.G., Porac, J.F. and Wade, J.B. (2004). Constructing deal networks: brokers as network 'architects' in the USIPO market and other examples. *Academy of Management Review*, **29**, pp. 50–72.
- Rider, C. (2009). Embedding inter-organizational relations in organizational members' prior education and employment networks. Working paper, Emory University.
- Sahlman, W.A. (1990). The structure and governance of venture-capital organizations. *Journal of Financial Economics*, **27**, pp. 473–521.
- Schmidt, K.M. (2003). Convertible securities and venture capital finance. *Journal of Finance*, **58**, pp. 1139–1166.
- Song, W.L. (2004). Competition and coalition among underwriters: the decision to join a syndicate. *Journal of Finance*, **59**, pp. 2421–2444.
- Sorenson, O. and Stuart, T. (2008). Bringing the context back in: settings and the search for syndicate partners in venture capital investment networks. *Administrative Science Quarterly*, **53**, pp. 266–294.
- Sorenson, O. and Stuart, T.E. (2001). Syndication networks and the spatial distribution of venture capital investments. *American Journal of Sociology*, **106**, pp. 1546–1588.
- Stuart, T.E. (2000). Interorganizational alliances and the performance of firms: a study of growth and innovation rates in a high-technology industry. *Strategic Management Journal*, **21**, pp. 791–811.
- Stuart, T.E., Hoang, H. and Hybels, R.C. (1999). Inter-organizational endorsements and the performance of entrepreneurial ventures. *Administrative Science Quarterly*, **44**, pp. 315–349.
- Tian, X. (2011). The causes and consequences of venture capital stage financing. *Journal of Financial Economics*, **101**, pp. 132–159.
- Tian, X. (forthcoming). The role of venture capital syndication in value creation for entrepreneurial firms. *Review of Finance*, forthcoming.

- Trapido, D. (2007). Competitive embeddedness and the emergence of interfirm cooperation. *Social Forces*, **86**, pp. 165–191.
- Tykvová, T. (2007). What do economists tell us about venture capital contracts? *Journal of Economic Surveys*, **21**, pp. 65–89.
- Tykvová, T. (2007). Who chooses whom? Syndication, skills and reputation. *Review of Financial Economics*, **16**, pp. 5–28.
- Ueda, M. (2004). Banks versus venture capital: project evaluation, screening, and expropriation. *Journal of Finance*, **59**, pp. 601–621.
- Verwaal, E., Bruining, H., Wright, M., Manigart, S. and Lockett, A. (2010). Resources access needs and capabilities as mediators of the relationship between VC firm size and syndication. *Small Business Economics*, **34**, pp. 277–291.
- Walker, G. (2008). The rise of ecommerce as an epidemic in the small world of venture capital. *Network Strategy*, **25**, pp. 3–29.
- Walske, J., Smith-Doerr, L. and Zacharakis, A. (2007). Effect of venture capital syndication networks on entrepreneurial success. *Frontiers of Entrepreneurship Research*, **27**, pp. 38–52.
- Wilson, R. (1968). The theory of syndicates. *Econometrica*, **36**, pp. 119–132.
- Wright, M. and Lockett, A. (2003). The structure and management of alliances: syndication in the venture capital industry. *Journal of Management Studies*, **40**, pp. 2073–2102.